

www.3aayaam.in

Section 11

Additional Info





Agenda

Best Practice

Limits

Common Challenges

Best Practices

- RA3 node types.
- Networking setup. Private Subnet, VPC Endpoint for S3.
- Proper IAM Roles.
- Implement security (RBAC, RLS, CLS, Audit Logging).
- Database Connections. Should not be more for OLAP.
- Distribution Key and Sort Key.
- Compression.
- Primary and Foreign Key have to be defined.
- Transactions.
- Single COPY command to load multiple files into a table.
- ALWAYS Archive!!

Best Practices

- Monitor (CloudWatch, System Views). Strategy.
- Leverage Zero-ETL to move data from OLTP to OLAP.
- Don't load data unless absolutely needed.
- Use Federated Query and Spectrum.
- Regular table maintenance (even though there is auto vacuum and auto analyze).
- Materialized View to better query performance.
- Streaming ingestion ???
- Proactive query tuning (monitor using SYSTEM tables/views).
- Always use WLM (Auto). Query Monitoring Rule.
- Serverless use case.

Redshift Limits (Provisioned)

- Concurrency level (query slots) for all user-defined manual WLM queues – 50 Hard limit. How to handle more concurrency?
- Concurrency scaling clusters – 10 (default). Soft Limit, it can be increased.
- No. of databases – 60. Hard Limit.
- Max connections for RA3 nodes – 2000. Hard Limit.
- Max connections for DC2 nodes – 500 to 2000. Hard Limit.
- Schemas per database – 9,900. Hard Limit.
- Max tables large, xlarge and xlplus (single-node) node types – 9,900. Hard Limit.
- Max tables xlplus (multi-node) node types – 20,000. Hard Limit.
- Max tables 4xlarge, 8xlarge, 16xlarge node types – 200,000. Hard Limit.

Redshift Limits (Serverless)

- Max connections – 2000. Hard Limit.
- No. of databases – 100. Hard Limit.
- Schemas per database – 9,900. Hard Limit.
- Max tables – 200,000. Hard Limit.

Challenges (to look out for)

- IAM roles.
- RLS policies and conflict.
- Resizing clusters.
- Loading JSON data in SUPER data type.
- Query running for a long time. Have to terminate.
- Setting DMS and making it up and running can be challenging owing to DMS's configuration options.
- MV not getting refresh incrementally. Its full refresh always.
- Streaming ingestion with MSK and Kinesis.
- Query tuning!!

Challenges (to look out for)

- Concurrent connections.
- Very large and complex EXPLAIN plan.
- Tuning for all queries!
- Altering DISTKEY and SORTKEY.
- Queries not getting directed to proper WLM queue.
- Concurrency level error (wlm_query_slot_count).
- Streaming ingestion MV cannot set to an earlier offset after a failure.

Redshift Concluded!

Thank you

www.3aayaam.in

